

How Much Syntactic Structure Have Cartographic Hierarchies Overlooked?

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Mapping Word Order: A Cartographic Perspective on Syntax
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1.

Setting aside the fact that there can be no precise answer to the question posed by the title of this talk, we can note in passing that any discussion of word order must keep silent elements in mind, both the antecedentless kind and the kind for which an antecedent is essential, whether via movement or via deletion (which may itself necessarily involve movement).¹

2.

Let me begin with a fairly familiar type of case. American English allows both of the following:

(1) You haven't finished your paper. I have *so* (finished it).

(2) You haven't finished your paper. I have *too* (finished it).

with no obvious difference in interpretation (though the example with *too* is more colloquial). One might initially think that *so* and *too* in such cases are just two variants of the same thing (though that would be incompatible with antisynonymy).²

However, *so* and *too* here can cooccur, to some degree of acceptability:

(3) ?I have *so too* (finished it).

but only in that order:

(4) *I have *too so* (finished it).

A cartographic conclusion might then well be that this *so* is in a higher projection than this *too*.

This would leave open, in a familiar way, at least the following questions:

(5) Why is *so* the higher one? Why not the reverse?³

(6) What is the link between this *so* and various others? Ditto for *too*.

Assume, following antihomophony (whether taken to be a principle or a heuristic),⁴ that the *too* under discussion is and must be the same element as the *too* of (7):

(7) They like phonology, *too*.

¹Cf. Kayne (2022).

²On antisynonymy, cf. Kayne (2024).

³This pair of questions arises in a very similar way for Cinque's (2015) placing of 'diminutive' above 'endearment'.

⁴On antihomophony, cf. Johns (1992, 84), Embick (2003, 156), Bobaljik (2012, 35) and Kayne (2019a).

Along the lines of Kayne (1998), the *too* of (7) is an attractor and can have *they like phonology* in its Spec, via movement, starting from a structure in which *too* is higher, as in:

(8) *too* [they like phonology]

If we now return to (the more compact variant of) (3), the possibility arises of a derivation having at an early stage:

(9) *too* [I have *so*]

with movement/attraction then yielding:

(10) [I have *so*] *too*

in which *I have so* is in Spec,*too*.

If (9) is correct, then the earlier tentative conclusion based on (3) vs. (4), which gave the impression that *so* is cartographically higher than *too*, is incorrect. In addition, we now see that the paired questions of (5) were not to the point.

The answer to the ordering question posed by (3) vs. (4) is arguably of a different sort entirely: The fact that *so* precedes *too* is a subcase of a more general fact about *too*, namely that *too* always follows whatever is focally aligned with it, one pair of relevant examples being:

(11) The fact that they like phonology, *too* is not surprising.

(12) *The fact that *too*, they like phonology is not surprising.

which fit well with the fact that in:

(13) John, *too*, likes phonology.

John can be in focus, but *phonology* cannot be, nor can *likes*. (The proposal just made rests on the assumption that in (3) *so* is focally aligned with *too*, which is plausible, but in need of further clarification.)

3.

In addition to (1) and (2), English allows, with strong stress on *yes* (indicated by underlining):

(14) You haven't finished your paper. Yes I have.

This *yes* cannot, to my ear, be embedded:

(15) You haven't finished your paper. *But I'm telling you that yes I have.

Nor can it cooccur with other CP elements, even when not embedded:

(16) You haven't spoken to Mary. (But) *To Mary/her yes I have.

which makes it natural to think that this *yes* is somewhere in the Comp area, in Rizzi's (1997) terms.

Attributing that kind of position to *yes* in (14) leaves open, as usual, the question whether it reached that position via external merge, or via internal merge/movement, as do focussed phrases found in the left periphery, and perhaps as do topicalized phrases in the same left periphery. Closer to left-peripheral *yes* may be *if* and its Catalan counterpart that Rigau (1984) argued not to be a complementizer in Bresnan's (1970) sense,⁵ but rather to be raised up from a lower position, much as Kayne (2010) later argued for English *that*.

⁵Extended to *if*, this will require some revision of Kayne's (1991) account of *We're not sure whether/*if to leave or not* and related facts in Romance, as well as a rethinking of why French *si* ('if') does not license 'stylistic inversion' - cf. Kayne and Pollock (2001).)

The idea that the *yes* in question obligatorily moves up from a lower position may be supported by its incompatibility with the earlier discussed *so* and *too*:

(17) You didn't finish your paper. Yes I did (**so*/**too*).
though one wonders whether or not (17) can be unified with the fact that this *yes* is hardly compatible with other post-finite-auxiliary material:

(18) You haven't been here recently. Yes I have (??*been*).

(19) You haven't been here recently. *Yes I've been.

(20) You haven't finished your paper. Yes I have (**finished* it).

To be noted is the contrast between (18) and the following, without strong stress on *yes*:

(21) Have you been here recently? Yes, I have (*been*).

This contrast may be related to:

(22) Have you been here recently? Yes.

(23) You haven't been here recently. *Yes.

(with (23) contrasting with French *si*).

4.

The case just discussed of *so* and *too* has something in common with that of *whole* and *entire*, which seem to yield identical interpretations in:

(24) We spent the whole day on just that one question.

(25) We spent the entire day on just that one question.

Antisynonymy would again say that *whole* and *entire* cannot be perfect synonyms, i.e. cannot be phonological variants of one element. And again we find that they can cooccur, with a clear preference for one order:

(26) We spent the whole entire day on just that one question.

(27) *?We spent the entire whole day on just that one question.

As before, we need to ask:

(28) Why is *whole* the higher one? Why not the reverse?

Part of the answer, differently from the case of *so* and *too*, may be:

(29) *Whole* is closer to *all* than *entire* is.

thinking of the fact that *all* can be high in the DP, to judge by its usual pre-article position. There may in turn be a link between (29) and the fact that *entire* (though not *entirely*, when modifying an adjective) needs an overt noun, while *whole* does not:

(30) On the whole/*entire, we agree.

(31) That's the whole/*entire of it.

just as *all* does not (in certain cases):

(32) They have five children. ?All/Not all are linguists. All five are linguists.

In addition, English allows (with sarcasm):

(33) They have all *(of) two students in their seminar this semester.

which can be transposed to *whole* to some extent:

(34) ?They have a whole two students in their seminar this semester.

but not at all to *entire*:

(35) *They have an entire two students in their seminar this semester.

5.

Sometimes, though, there appears to be no ordering preference, where we might cartographically have expected one:

(36) He's probably really made a bad mistake.

(37) He's really probably made a bad mistake.

There is, on the other hand, a difference in interpretation, which I suspect is best understood by taking *really* in (37) (and perhaps *probably* in (36)) to be associated with a silent counterpart of *true* (capitals will indicate silence):

(38) he has really TRUE probably...

with a derivation that crucially has at an earlier stage:

(39) really TRUE he has probably...

where 'he has' moves past 'really TRUE', probably via remnant movement. If so, then in (37) *really* will be part of the cartographic hierarchy associated with TRUE, rather than the one associated with *made*.

6.

A silent matrix predicate may also be appropriate in a similar way for sentences like:

(40) But he didn't say that at all.

which readily has the interpretation of:

(41) But it's not at all true that he said that.

suggesting that the *n't* and *at all* of (40) might actually go with a silent matrix predicate (BE) TRUE,⁶ rather than with the pronounced verb *say*, with as a first approximation a derivation like (omitting traces and *do*-support):

(42) ...n't at all TRUE he said that —>

...n't [he said that] at all TRUE —>

...he n't [said that] at all TRUE

in which both negation and *at all* go with TRUE and not with *say*, contrary to (cartographic) appearances.

7.

On negation in a non-obvious matrix sentence, we might also think of the following contrast:

(43) Not all the students got good grades.

(44) *We gave good grades to not all the students.

which suggests, for (43):⁷

(45) IT IS not THE CASE THAT all the students got good grades

(Why *not a single* acts differently from *not all* remains to be understood.)

8.

Negation in a matrix sentence is arguably exactly what is found in Jacalteco, a Mayan language discussed by Craig (1977, 26), who says: "For the negation of sentences with a finite verb, the negation word *mat* combines with the existential *ay*, yielding *mach*. The phonological derivation of *mach* from *mat* + *ay* is not very clear, but the fact that it has occurred is obvious in the negation of possessive expressions:

⁶Cf. Cinque (1976), Zanuttini (1997) and Colombini (2007, 74) on Italian *mica* and dialect counterparts.

⁷Cf. Klima (1964, 300) on neg as S-initial; also McCawley's (1973) taking *not* to be a verb.

- (46) a. ay hin melyu ('exist my money')
 b. *mat ay hin melyu
 c. mach hin melyu (another possibility: mach = mat+oj')

If Craig is right, there's an existential *ay* in every negative sentence in Jacaltec. As if Jacaltec expressed, for example, *You don't like coffee* as ??*You're liking coffee doesn't exist*. (Cf. English *It's not the case that you like coffee*.)

9.

The above recalls:⁸

- (47) The flowers are in the kitchen on purpose.

in which *on purpose* is an adverbial that is almost certainly best understood as being (cartographically) associated with a silent matrix predicate CAUSE rather than with overt *are*.

10.

In the same spirit, consider:

- (48) If only we had worked harder!

in which *only* seems untethered, unless we take (48) to be as in:

- (49) if only IT WERE THE CASE THAT we had worked harder

11.

A silent matrix predicate may also be called for in cases of focus-preposing, in a way that would bring focus-preposing close to cleft sentences. That focus preposing might consistently involve a bisentential structure is suggested by what Ofori (2011) said about Akan: "The basic focus marker, we argue, is an equational copula/verb that joins a focused-unit and a presuppositional-constituent".⁹

Anna Szabolcsi (p.c.) has made a similar suggestion (based on work by Soo-Hwan Lee¹⁰) regarding Korean NOM(inative)-marked items, noting that if focussed NOM-marked phrases were actually parts of clefts, then the seemingly dual role of NOM in Korean (as Case-marker and as focus-marker) would dissolve and become compatible with antihomophony.

Guglielmo Cinque (p.c.) has noted that Italian focus-preposing and Italian clefts differ with respect to negative phrases, which can be subject to focus-preposing, but cannot be clefted. However, Pollock (2021) notes that French interrogatives having the form of interrogative clefts, as in:

- (50) Qu'est-ce que tu fais? ('what is it that you are-doing')

"despite their bi-clausal forms would best be rendered as simple mono-clausal Wh-questions in English...none have a cleft interpretation". Thus it might be that focus-preposing is akin to French (50) in that respect.

The question why focus-preposing would require a (perhaps silent) copula feels close to the question of:

- (51) Mary *(is) a good teacher/Ann's best friend.

⁸Cf. Kayne (2023).

⁹Cf. also Titov (2019) and Pearce (2021).

¹⁰Cf. Lee (2024).

If a (sometimes silent) copula is universally required in sentences like (51) for reasons of interpretation, that requirement might well carry over to the pair consisting of a focussed phrase and the sentential constituent that appears to be its sister phrase.¹¹

12.

In a range of cases, then, cartographic (and other syntactic) work will want to be on the lookout for additional syntactic structure, with more silent elements, and more intricate derivations than might have met the eye, at first glance.

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¹¹Relevant here is Moro (1997), as well as den Dikken (2006).

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